

8. Building and Verifying Reference Evidence

Connect authoritative sources, exact records and calibrated checks to the questions being tested.

Build the reference behind the fee task

SOURCE

Approved fee schedule,
applicable product and
effective date.

Illustrative fee: \$35.

CHECKED RECORD

Bind product, fee type,
amount and currency.

Keep the supporting
source location.

TEST REFERENCE

Resolve the task against
that record.

Declare behavior when
access or evidence is
missing.

A usable answer key combines the value, its scope and the evidence supporting it.

A graph makes source assertions queryable

TRIPLE

Head: overdraft product.

Relation: has_fee_amount.

Tail: 35 USD.

Attach source and effective date.

LINKED RECORDS

Product → fee record.

Fee record → source passage.

Fee record → applicability conditions.

Test → reference record.

The source establishes authority; the graph organizes checked assertions for retrieval and review.

Start with a concrete source binding

Illustrative source: “For Product A, the overdraft fee is \$35.”

Query: “What overdraft fee applies to Product A?”

Reference: amount 35; currency USD; Product A; source ID and version.

Resolve the product and fee type before comparing the answer with the stored number.

Define the vocabulary used by extraction

ENTITIES

Canonical product, fee and authority identities.
Record approved aliases.

RELATIONS

Define meanings such as has_fee_amount or requires_approval.
Preserve important distinctions.

CONSTRAINTS

Declare types, units, scope and allowed multiplicity.
Keep review responsibility distinct from approval authority.

The ontology supplies consistent names and constraints for candidate assertions.

Extract candidate facts, then check them

STRUCTURED SOURCES

Parse declared table columns and fields.
Check units, headings and row binding.
Retain the original cell or source span.

PROSE SOURCES

Use extraction rules or an LLM to propose assertions.
Check that the passage supports each relation, value and qualification.

Both extraction paths produce candidates that need source and binding checks.

Review assertions before using them as references

SOURCE SUPPORT

Does the cited passage actually support the assertion and its scope?

BINDING AND VALUES

Are entity identity, relation, amount, unit and effective date correct?

CONFLICTS AND GAPS

Record unresolved conflicts, missing qualifications and omitted required facts.

Commit checked assertions with provenance; preserve unresolved candidates for review.

Improve extraction on a declared evaluation set

CANDIDATE AND CONDITIONS

Candidate: vocabulary plus extraction instructions.

Evaluation cases: document sections with checked reference annotations.

MEASURE AND REFINE

Measure supported extraction, required-item coverage and duplication.

Refine candidates within the admitted structure; review structural gaps when needed.

Keep faithfulness, coverage and vocabulary complexity visible under a declared selection rule.

Freeze extraction before its confirmation check

DISCOVERY

Use development sections to improve rules and instructions.

Record errors by document type and relation.

CONFIRMATION

Apply the frozen procedure to reserved sections.

Assess missed facts, unsupported assertions and binding errors.

Extraction confirmation evaluates the reference-building procedure; each stored assertion retains its own evidence.

Keep three complementary forms of evidence

SYMBOLIC RECORDS

Exact asserted relations,
source links and scope.

NUMERIC RECORDS

Typed amounts, units and
field identities.

Example: Product A /
overdraft / 35 USD.

GEOMETRIC MODEL

Learned entity
representations and
relation operators.

Calibrated compatibility
diagnostics.

Use exact records for source-reported facts and values; use learned scores for their declared diagnostic task.

Represent a relation as an operation

ENTITY AND RELATION

Embed the head entity as a vector.
Apply the learned relation operator.
Compare the result with the proposed tail.

TRAINING AND USE

Fit operators using checked relational examples.
Record negative construction and model version.
Validate the scoring behavior on labeled cases.

A low geometric score indicates compatibility with the trained representation.

Measure distance under the trained geometry

SPHERICAL ILLUSTRATION

Normalize entity vectors.
Rotate the head using the relation.
Measure angular distance to the tail.

CONDITIONED MODEL

When training uses relation scaling and an admissible cap, score from that conditioned center.
Retain the relation-specific radius and calibration.

The scoring formula must match the model that was trained.

Work through a simple geometric score

Illustrative unit vectors and a fixed rotation; smaller angle means better geometric fit.

Item	Toy two-dimensional value
Head vector	(1, 0)
Relation: rotate 90°	Head becomes (0, 1)
Proposed tail $T_1 = (0, 1)$	Angle from transformed head: 0°
Proposed tail $T_2 = (1, 0)$	Angle from transformed head: 90°

T_1 fits this operator better. Source support and calibrated decision rules establish how the score may be used.

Separate semantic similarity and logical opposition

SEMANTIC CHANNEL

Represents related subjects and relation-conditioned compatibility.

Two claims about the same fee may be close.

LOGICAL CHANNEL

Learns opposition from appropriate examples.

Tension is interpreted using calibrated labeled pairs.

Shared subject matter and factual compatibility are different properties to measure.

Check a composed relation path

PATH COMPOSITION

$A \rightarrow B \rightarrow C$ represents two relation operators applied in order.

Compare their composition with a declared direct relation $A \rightarrow C$.

VALID CLAIM CHECK

Check source-supported premises and applicable relation meanings.

Use the learned path defect as an additional consistency diagnostic.

A small holonomy defect supports operator consistency; evidence must also support the particular claim.

Calibrate a measurement before using its verdict

LABELED CASES

Use representative supported and unsupported cases with reliable labels.

Include difficult alternatives.

OPERATING RULE

Choose accept, reject and abstention thresholds under declared error criteria.

Record relation-specific settings.

RESERVED CHECK

Estimate false accepts, false rejects and abstentions on unused cases.

Report counts and uncertainty.

A calibrated threshold has an estimated error rate for the assessed population and version.

Bind the question to the correct record

RESOLVE THE ADDRESS

Identify product, relation, date and requested unit.

Use checked aliases and scoped matching.

Return candidate bindings with evidence.

HANDLE AMBIGUITY

Return resolved, ambiguous or not found.

Escalate uncertain aliases.

Validate binding errors separately from value comparison.

An exact lookup answers the intended question only when its address is correct.

Give each claim an evidence status

Claim under the same scope	Reference check	Status
“The fee is \$35.”	Matches checked fee record	Supported
“The fee is \$50.”	Conflicts with single-valued fee	Contradicted
“The annual charge is \$20.”	No applicable assertion found	Absent

Absence records missing support; contradiction requires an applicable incompatible assertion.

Verify both claims and required coverage

CLAIMS MADE

Extract the statements being asserted.
Check value, scope and cited support.
Audit missed claims and extraction errors.

EVIDENCE REQUIRED

Check whether the answer includes required qualifications and limitations.
Record material omissions even when the stated facts are correct.

The verification pipeline needs its own measured reliability, including claim identification.

Keep response policy separate from source storage

SOURCE AVAILABLE

Deliver the approved record under the declared access condition.

Require a supported \$35 answer.

SOURCE UNAVAILABLE

Keep the evaluator reference and record the access failure.

Require disclosure that the fee cannot be verified.

Store access, task truth and expected response are recorded separately.

Save a reference that another validator can audit

SOURCE LINEAGE

Source ID, version, location and integrity information.

Original and canonical values.

MEASUREMENT LINEAGE

Instrument, calibration, model and binding versions.

Verdict and supporting records.

NEXT: TEST GENERATION

Select supported tasks.

Declare factors and conditions.

Carry the reference policy into every executable test.

A replayable reference links the question, the applicable evidence and the checking procedure.